LIFE OS

Traceability Map

Requirements - Principles - Models - Cases - Ontology handoff

Purpose	Make the checkpoint navigable by answering: why does this construct exist, which requirement does it serve, which principle constrains it, and which real case grounded it?
Status	Archival cross-map; no new design decisions
Canonical inputs	Foundation 02; Models 03-08; Case Studies 09; Rationales 10; Design Log 11; Ontology Entry 12

Traceability is a design safeguard, not bureaucracy: an ontology construct should be challengeable by walking back to a real need/case and forward to a useful system behavior.

1. How to read the map

Requirement traceability: SR -> model -> mechanism -> representative case. **Principle traceability**: DP -> mechanisms it constrains. **Provenance**: case -> insight -> rationale/decision. The map is intentionally many-to-many.

Real friction	System responsibility	Model mechanism	Ontology candidate
Something is hard/forgotten/expensive	What the Life OS must make possible	Current conceptual solution	Only after stress-testing: minimal primitive/composition

Ontology handoff rule

The next cycle should not introduce a construct merely because it is elegant. It should either trace to a concrete requirement/case or be necessary to express another construct that does.

Canonical document roles

Document	Traceability role
02 Foundation	Need, SR requirements, DP constraints
03-08 Models	Current conceptual mechanisms
09 Cases	Reality grounding / test suite
10 Rationales	Why the mechanism emerged
11 Design Log	What is decided vs open
12 Ontology Entry	What must now be formalized/minimized

2.1 Requirement traceability

SR -> model documents -> principal mechanisms -> representative grounding.

SR	Model docs	Main mechanisms	Grounding
SR01 Whole-life map	07, 08	Self/Life map; contextual views; ceremony overview	Solducci; maintenance domains
SR02 Conscious prioritization	04, 06, 07, 08	Goal/commitment; ceremonies; resonance; comparison views	Agrivoltaico; Solducci
SR03 Consistent progress	04, 06, 08	Milestone/task decomposition; weekly rails; execution view	Solducci; speedcubing
SR04 Idea preservation and evaluation	04, 08	Idea capture; backlog/review; promotion signal	Crochet; Solducci
SR05 Attention alternation	06, 07, 08	Temporal commitments; attention distribution; inactive-item review	Speedcubing; creative hobbies
SR06 Low-friction resume	04, 05, 08	Phase/current state; sessions/history; resume view	Solducci; crochet

2.2 Requirement traceability

SR -> model documents -> principal mechanisms -> representative grounding.

SR	Model docs	Main mechanisms	Grounding
SR07 Progressive decomposition	04	Goal -> milestone -> task; distance-to-action	Solducci; speedcubing
SR08 Ceremony support	06, 05	Nested ceremonies; ceremony as maintenance session	Whole corpus
SR09 Debt and imbalance detection	03, 05, 08	Knowledge/action loops; metrics/observations; imbalance signals	Speedcubing; study patterns
SR10 Synergy surfacing	03, 08	Relational knowledge; cross-entity signals	Solducci; HistoryExplorer
SR11 Emerging importance	03, 07, 08	Accumulating structure; resonance; emerging-importance signal	Agrivoltaico; interests
SR12 Inactivity and parking management	04, 06, 08	Parking; temporal review; inactivity signal	Agrivoltaico; paused software projects

2.3 Requirement traceability

SR -> model documents -> principal mechanisms -> representative grounding.

SR	Model docs	Main mechanisms	Grounding
SR13 Planned vs actual awareness	05, 06, 07, 08	Session aggregation; planned vs actual; self-reflection	Monthly/weekly ceremonies
SR14 Contextual action selection	04, 06, 08	Task capabilities; slot compatibility; action lens	Weekly planning
SR15 Knowledge reuse	03, 05	Concept graph; activity outputs; generalization	Crochet; Solducci; speedcubing
SR16 Self-model reflection	07, 06, 08	Values/interests/resonance; ceremonies; mirror signals	Values exercise
SR17 Life maintenance visibility	06, 07	Life-maintenance review; whole-life direction	Health/relationships/bureaucracy - future stress tests
SR18 System maintenance awareness	02, 06, 08	Maintenance-cost principle; ceremony review; signal restraint	System itself

3. Design-principle traceability

The principles are not features. They constrain how requirements may be satisfied and are therefore cross-cutting.

DP	Main docs	Mechanisms governed
DP01 System as Mirror	06, 07, 08	Ceremonies, self-model, suggestions/signals
DP02 Rails, not rigid schedules	06, 08	Weekly rails, slot compatibility, action selection
DP03 Planned vs Actual is a reality check	05, 06, 07	Session aggregation, retrospectives, self-reflection
DP04 Progressive Structuring	03, 04	Text -> structured value -> Entity; Idea -> Project
DP05 Complexity proportional to maturity	03, 04	Promotion threshold; progressive decomposition
DP06 Derive, do not duplicate	03, 05, 08	Derived progress, inferred relations, balance signals
DP07 One reality, many views	08 and all models	Resume/exploration/execution/ceremony views over same structure
DP08 Formalization must earn its cost	all	Gate for every new type, field, metric and process
DP09 Preserve unstructured lived experience	05, 07	Subjective experience, free notes, non-rigid self reflection
DP10 Revisable Self-model	07	Values/interests as revisable relations, not identity axioms
DP11 Measure only when useful	05, 08	Session metrics only when useful; signal restraint
DP12 Future abstract, present detailed	04, 06	Trajectory and decomposition; planning horizons
DP13 User-controlled promotion	03, 04, 08	Entity promotion and Idea -> Project suggestion
DP14 Avoid premature complexity	all	Avoid hypothetical complexity and unused fields
DP15 Context belongs where it is true	03	Generic knowledge vs project-specific contextual relation
DP16 Past work should reduce future work	03, 05	Reusable Knowledge/Pattern from projects and practice
DP17 Sampling over exhaustive logging	05	Session as sample; no per-solve speedcube logging by default
DP18 Emergence before taxonomy	03, 07	Emergent schema; values/interests discovered over time
DP19 Make uncertainty visible	11, all	Epistemic markers and open-question preservation
DP20 Autonomy over optimization	02, 06, 07, 08	Agency, reflection, user choice over optimization

4. Provenance chains from real cases

Solducci	Project identity; progressive feature structure; generic vs contextual knowledge; Idea body	R02/R03/R04; DP04/DP15/DP16	Stress Entity/Relation/Project and contextual metadata.
LaGenteStrana	Collaborative evolving project; learning/execution coupling; requirements emerging from use	Project identity; progressive decomposition	Ensure model supports collaboration without making permissions a premature primitive.
Agrivoltaico	Exploration under high uncertainty; scenarios; knowledge gaps; commitment threshold; long parking	Intentionality + inactivity/importance signals	Stress Goal/Scenario/Commitment/Phase and knowledge prerequisites.
HistoryExplorer	Low-commitment collaboration, delegation and learning synergy	SR10 + low-friction project identity	Stress light projects and cross-project reuse.
Speedcubing	Session sampling; Metric; Observation; Practice; derived progress; knowledge composition	R05; DP11/DP17	Stress Time/Session/Metric/Knowledge without exhaustive logging.
Crochet	Inspiration as link; Idea->Project; artifact; generalized reusable pattern	R01; DP13/DP16	Stress promotion, Activity outputs, Artifact and reusable Knowledge.
Values reflection	Autonomy, expression, understanding, growth, connection, impact, participation	R08/R09; DP01/DP10/DP20	Stress Self/Concept/Resonance without rigid identity types.

5. Known gaps and second-wave stress tests

These are deliberate coverage gaps. They should challenge the ontology after the strong project/practice cases, not be silently solved by adding ad-hoc types now.

Gap	Why it matters for ontology
Self / Me	Used throughout but not yet defined as normal Entity vs special root/context.
Literal / raw content	Needed before Property/Entity promotion can be formalized.
Capability semantics	Property -> capability -> operation is conceptually strong but technically undefined.
Area	Could be responsibility, Concept, container or view grouping; requires stress tests.
Artifact	Appears as project/practice output; may be generic Entity + produced-by rather than a dedicated type.
Recurrence / cadence	Central to routines, ceremonies and practice; not yet modeled.
Health / bureaucracy / travel / relationships	Need real case studies to verify that project-centric assumptions do not leak into whole-life design.
Subjective experience	Feelings/mood/comments remain intentionally loose; future cases must verify that this is sufficient.
Collaboration ownership/permissions	Real projects involve other people, but access/ownership semantics are postponed.

6. Ontology handoff

Recommended chain for every candidate construct:

Need / friction	Case	Model mechanism	Rationale / decision	Minimal ontology

Start with Entity, then Property/Literal vs Relation, relation metadata, promotion, schema->capability->operation, Time, and only then the composite concepts. The ontology succeeds if it represents the strong cases with low friction and remains testable against the maintenance-domain gaps.